MINING DEALER BEST PRACTICE

Inversion of 793D A-Frame Pin

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

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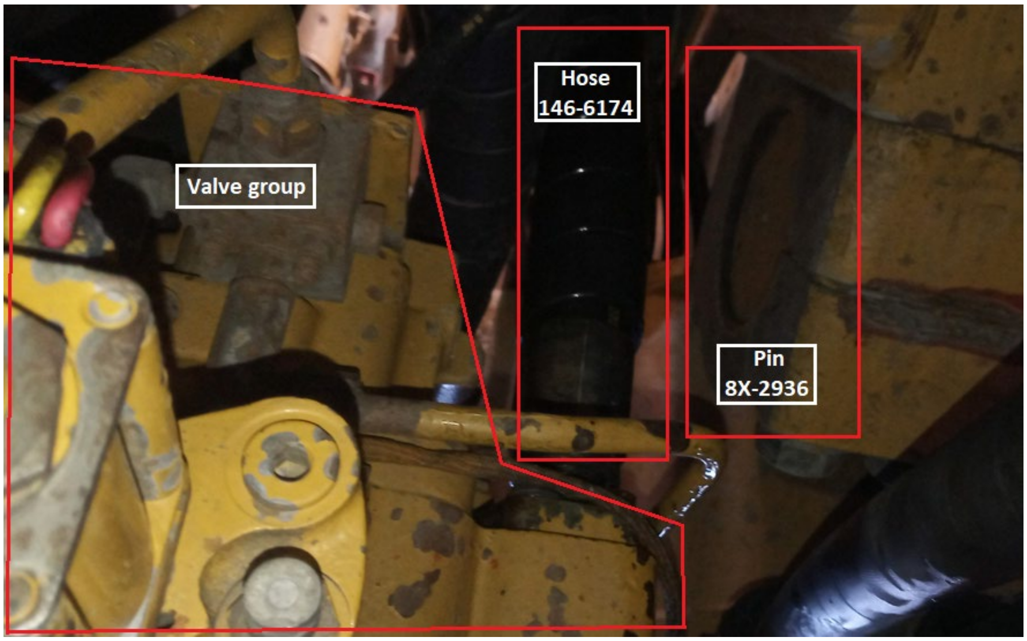
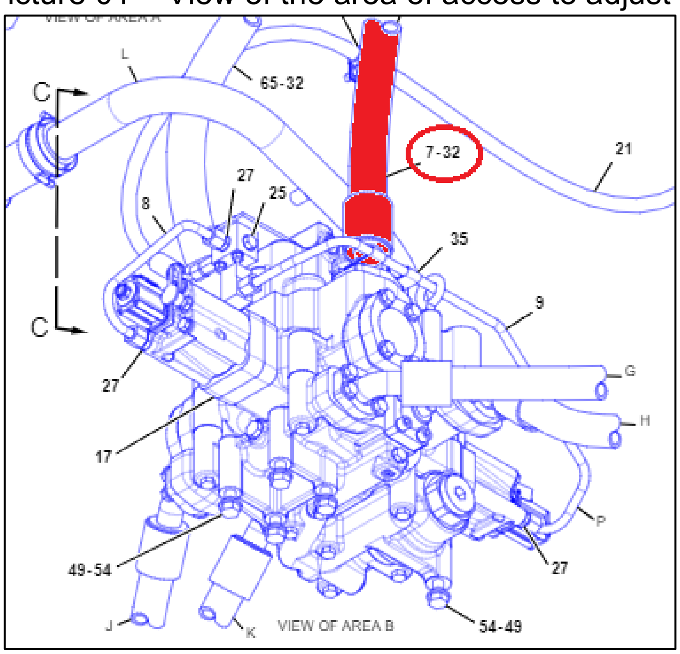
1.0 Introduction

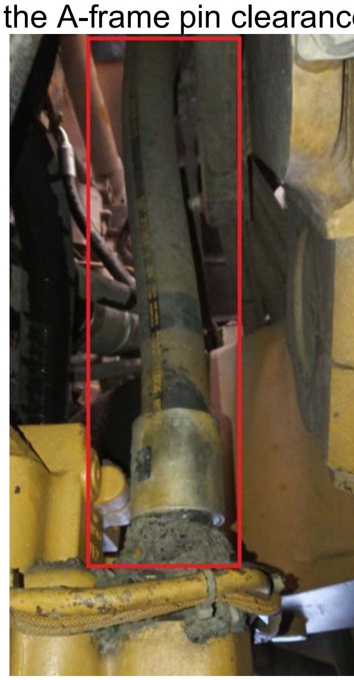
The procedure to adjust the 793D A-frame pin clearance (PN 8X-2936) requires removing the hoist system hose 146-6174 to gain access to the plate 8X-2939 and bolts 8T-0664 to perform the final torque.

The hose removal will expose the hydraulic system to external contaminants and will also increase the hours to execute the pin adjustment procedure.



Picture 01 – View of the area of access to adjust the A-frame pin clearance.



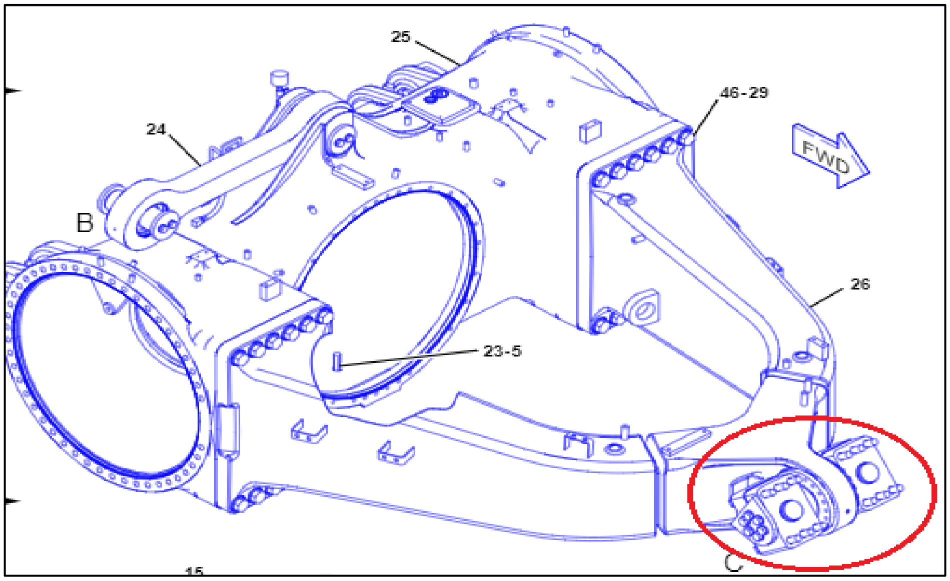


Pictures 02 & 03 – Detail of hoist system hose 146-6174 that needs to be removed to adjust the A-frame pin clearance.

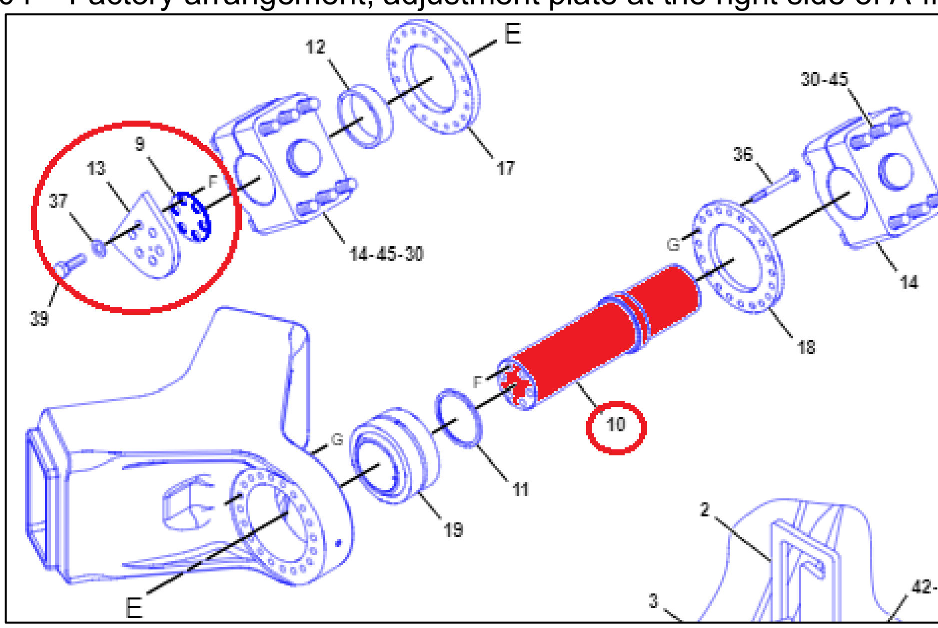
This Best Practice will depict the procedure to invert the A-frame pin thus eliminating the contamination risk as well as reducing labor hours and improving machine availability, with benefits for either customers or dealers.

2.0 Best Practice Description

The original installation position of the A-frame pin 8X-2936. The plate 8X-2939 and the adjustment shims 8X-2934 are on the right side next to the hoist system valve group.



Picture 04 – Factory arrangement, adjustment plate at the right side of A-frame pin.



Picture 05 – Factory arrangement, detail of bolts and adjustment plate of A-frame pin.

To avoid removing the hose to perform the pin adjustment, the customer changed the installation side of the pin. Now the adjustment plate is on the left side where there is enough space to carry out the pin clearance adjustment without the need to remove other parts.

Picture 06 – A-frame pin after inversion, adjustment plate at the left side of A-frame pin with free space for pin clearance adjustment.

Before inverting the A-frame pin, secure the A-frame front end using a hydraulic stand. It is suggested to use 9U-7489 100-ton hydraulic lift stand. Example of another supplier’s hydraulic lift stand is in the Picture 08. Use a suitable lifting device (not shown) to remove pin. The weight of the 8X-2936 A-frame pin is approximately 73kg.

1. Removal procedure

⚠ WARNING

Personal injury or death can result from improper assembly procedures.

Do not attempt any assembly until you have read and understand the assembly instructions.

⚠ WARNING

Personal injury or death can result from improper lifting or blocking.

When a hoist or jack is used to lift any part or component, stand clear of the area. Be sure the hoist or jack has the correct capacity to lift a component. Install blocks or stands before performance of any work under a heavy component.

Approximate weights of the components are shown. Clean all surfaces where parts are to be installed.

1.1. Remove the hydraulic lines to gain access to the A-frame pin as depicted in Picture 07.

Picture 07 – Hydraulic lines to be removed to gain access to the A-frame pin.

1.2. Remove the hoist valve group.
1.3. Remove the 12 bolts that attach the retaining caps of A-frame pin.
1.4. After removing the 12 bolts, support the rear end of differential housing with a hydraulic lift stand and raise it until the yoke contacts the lower face of the truck frame, thus allowing to remove and invert the A-frame pin. See Picture 08.

Picture 08 – Example of list stand to support the differential housing.

2. Installation procedure

Proceed in the reverse order of removal procedure.

3. General information

These procedures take 8 calendar hours. After doing so, the procedure to check A-frame pin end play decreases from 4 calendar hours to 2 calendar hours and the maintenance interval remains 2,000 hours. After executing the end play check 4 times (8,000 hours), customer obtain gains in machine downtime. For machine downtime dilution, it is suggested to execute the pin inversion during:

- the yoke rebore;
- the yoke weld repair; or
- the oversize bearing installation.

The 793D Operation and Maintenance Manual SEBU7792 states that inspecting the end play adjustment of the rear axle housing is recommended at the initial 2000 service hours. Yearly inspection and maintenance of the end play of the rear axle housing will maximize service life. However, in case of severe applications, the end play adjustment of the rear axle housing is recommended at every 6 months. Dealership and customer may use any of these maintenance intervals to invert the A-frame pin to mitigate impact in machine downtime.

3.0 Implementation Steps

1. Obtain the total labor hours required to adjust A-frame pin clearance in its original configuration, i.e. adjustment plate at the right side of A-frame pin.
2. Invert the pin according to the procedure shown in topic 2.0.
3. Execute the procedure to adjust A-frame pin clearance, i.e. adjustment plate at the left side of A-frame pin. Register the total labor hours and compare it with original configuration.

4.0 Benefits

- Reduces labor hours and consequently the machine downtime to adjust A-frame pin clearance – case study at the customer Kinross shows that labor hours reduced from 4 hours to 2 hours with one technician.
- Eliminate the source of contamination by avoiding the removal of the hoist system hose 146-6174 and oil refill.

5.0 Resources Required

This procedure requires the use of an 100 ton hydraulic lift stand, a torque wrench to tight the bolts, a micrometer and a flick to measure the shims. It is recommended to execute the inversion of A-frame pin during a bushing replacement.

6.0 Supporting Attachments / References

- Operation and Maintenance Manual, SEBU7792, “793D Off-Highway Truck.
- Testing and Adjusting, RENR8327, "793D Power Train Testing and Adjusting", "Rear Axle Housing End Play - Adjust".

7.0 Related Best Practices

Inversion of the body pivot pins on mining trucks (number 1.01-210211-1).

8.0 Acknowledgements

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